1. Create a temperature converter from Celsius to Fahrenheit.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="Enter Value to be Convert:"
        android:textSize="20dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/editTextNumberDecimal"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:hint="Enter Value"
        android:textSize="20dp"
        android:textStyle="bold"
        android:ems="10"
        android:inputType="numberDecimal"
        />

    <Button
        android:id="@+id/button"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text="Convert" />

    <TextView
```

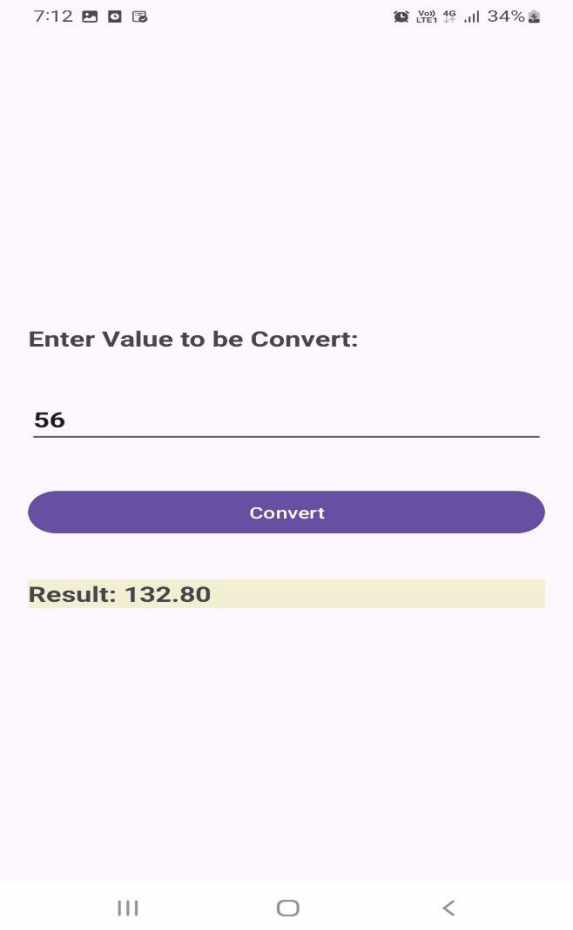
```
        android:id="@+id/txtRes"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:text=""
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="#F2EFD3"
        />
</LinearLayout>
```

JAVA CODE:

```
package com.example.appprno14;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;

 import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        EditText edt = findViewById(R.id.editTextNumberDecimal);
        TextView txtRes = findViewById(R.id.txtRes);
        Button btn = findViewById(R.id.button);


        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String strInput = edt.getText().toString();
                if (strInput.isEmpty()) {
```

```
            Toast.makeText(getApplicationContext(), "Enter Value", Toast.LENGTH_SHORT).show();
            return;
        }

        double dVal = Double.parseDouble(strInput);
        double dRes = 0;
        dRes  = (dVal * 9/5) + 32;
        txtRes.setText(String.format("Result: %.2f", dRes));
    }
  });
 }
}
```

OUTPUT:



2. Extend the app to convert Fahrenheit to Celsius.

3. Design a currency converter with fixed exchange rate.

4. Implement a volume converter (liters ↔ gallons).

5. Add a spinner to select conversion type and update UI accordingly.

6. Validate input and show error messages for invalid entries.

XML CODE:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    tools:context=".MainActivity">


    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="20dp"
        android:textSize="20dp"
        android:textStyle="bold"
        android:text="Select Conversion Type:"
        />

    <Spinner
        android:id="@+id/spinner"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
```

```
    android:textSize="20dp"
    android:textStyle="bold"
    android:layout_margin="20dp"
    />

<TextView
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
    android:text="Enter Value to be Convert:"
    android:textSize="20dp"
    android:textStyle="bold"
    />

<EditText
    android:id="@+id/editTextNumberDecimal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
    android:hint="Enter Value"
    android:textSize="20dp"
    android:textStyle="bold"
    android:ems="10"
    android:inputType="numberDecimal"
    />

<Button
    android:id="@+id/button"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
    android:text="Convert" />

<TextView
    android:id="@+id/txtRes"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="20dp"
```

```
        android:text=""
        android:textSize="20dp"
        android:textStyle="bold"
        android:background="#F2EFD3"
        />

</LinearLayout>
```

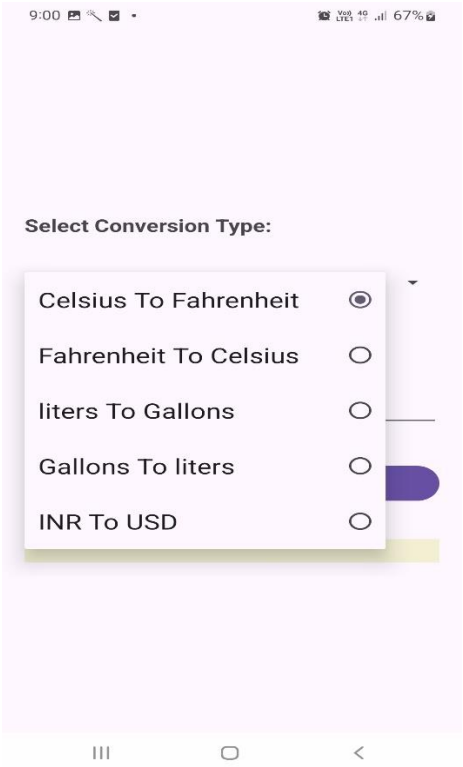
JAVA code:

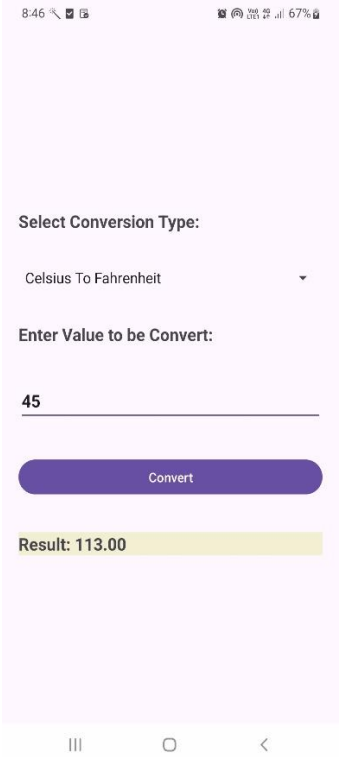
```
package com.example.appprno14;

import android.os.Bundle;
import android.view.View;
import android.widget.ArrayAdapter;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Spinner;
import android.widget.TextView;
import android.widget.Toast;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    String [] strTypes = {
            "Celsius To Fahrenheit",
            "Fahrenheit To Celsius",
            "liters To Gallons",
            "Gallons To liters",
            "INR ↔ USD"
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

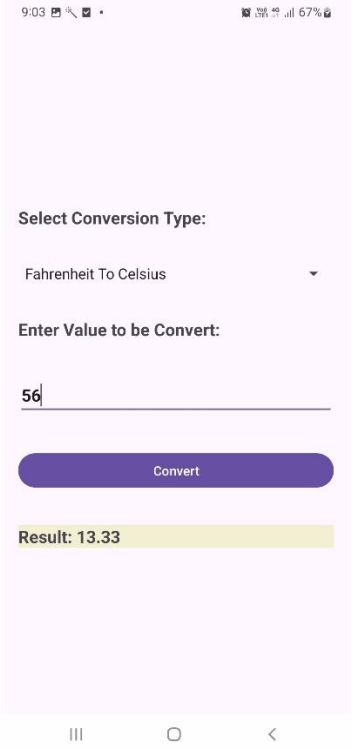
```
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        Spinner spin = findViewById(R.id.spinner);
        EditText edt = findViewById(R.id.editTextNumberDecimal);
        TextView txtRes = findViewById(R.id.txtRes);
        Button btn = findViewById(R.id.button);


        ArrayAdapter<String> adp = new ArrayAdapter<String>(this,
android.R.layout.simple_spinner_item, strTypes);

        adp.setDropDownViewResource(android.R.layout.select_dialog_singlechoice);
        spin.setAdapter(adp);

        btn.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                String strInput = edt.getText().toString();
                if (strInput.isEmpty()) {
                    Toast.makeText(getApplicationContext(), "Enter Value",
Toast.LENGTH_SHORT).show();
                    return;
                }

                double dVal = Double.parseDouble(strInput);
                double dRes = 0;
                String selected = spin.getSelectedItem().toString();

                switch (selected) {
                    case "Celsius To Fahrenheit":
                      dRes  = (dVal * 9/5) + 32;
                       break;

                    case "Fahrenheit To Celsius":
                       dRes = (dVal - 32) * 5/9;
                       break;
```

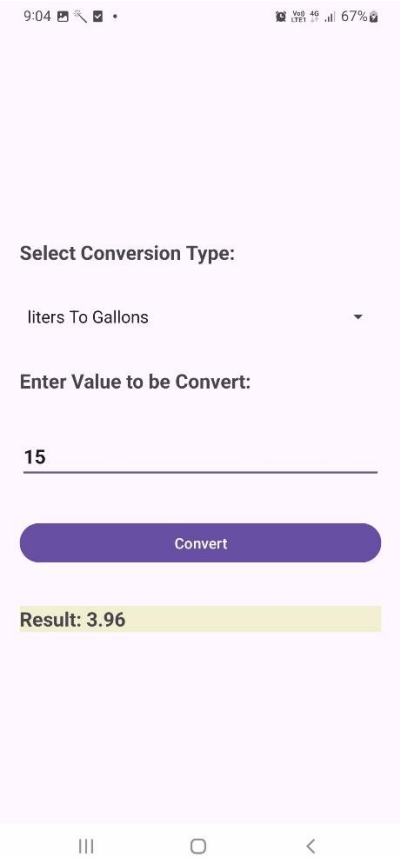
```
            case "liters To Gallons":
                dRes = dVal * 0.264172;
                break;

            case "Gallons To liters":
                dRes = dVal * 3.78541;
                break;

            case "INR To USD":
                dRes = dVal * 0.011;
                break;
        }
        txtRes.setText(String.format("Result: %.2f", dRes));
    }
  });
 }
}
```

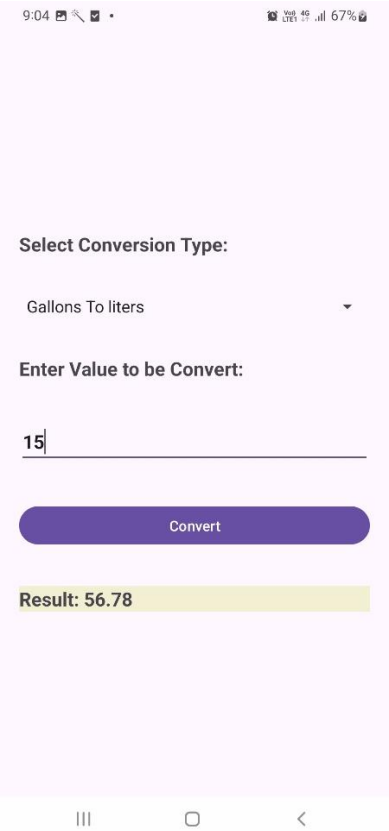
OUTPUT:

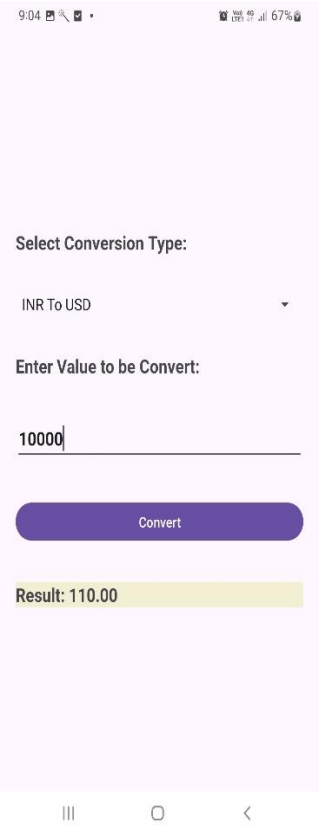







9:04
67%
Select Conversion Type:
liters To Gallons
Enter Value to be Convert:
15
Convert
Result: 3.96


9:04
67%
Select Conversion Type:
Gallons To liters
Enter Value to be Convert:
15
Convert
Result: 56.78


9:04
67%
Select Conversion Type:
INR To USD
Enter Value to be Convert:
10000
Convert
Result: 110.00